

PROJECT REQUIREMENT DOCUMENT

Final Capstone Project — Week 6

Project Name	Company Knowledge AI Assistant
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Organization	Petalnexus Pvt. Ltd.
Project Type	AI Automation / RAG Assistant

1. Problem Statement

Employees often need quick access to company policies, SOPs, FAQs, and internal documentation. When information is distributed across multiple documents or requires manual assistance, finding the correct information can take time and may result in inconsistent answers.

The project will provide a centralized AI-powered Company Knowledge Assistant that accepts natural-language questions and answers them using the organization's approved knowledge base. The assistant must avoid inventing information and clearly indicate when requested information is unavailable.

2. Current Process + Pain Points

Current Process

- Employees identify the policy, SOP, FAQ, or internal document relevant to their question.
- They manually search available documents or ask another person for assistance.
- The relevant information is read and interpreted manually.
- Questions requiring clarification or authority are referred to the appropriate person.

Pain Points

- Manual searching can be time-consuming.
- Information may be distributed across multiple documents.
- Relevant information can be difficult to locate quickly.
- Repeated questions create unnecessary workload for HR or responsible staff.
- Manually provided answers may be inconsistent.
- An AI assistant could provide incorrect information if it answers from general knowledge instead of approved company sources.

3. Proposed Solution Outline

The proposed solution is a Company Knowledge AI Assistant based on Retrieval-Augmented Generation (RAG). Approved company policies, SOPs, FAQs, and internal documents will be processed, converted into embeddings, and stored in a vector database. When a user asks a question, the system retrieves relevant knowledge and uses it as context for generating a grounded response.

- Ingest approved company knowledge from documents and structured sources.
- Extract, clean, and split content into manageable chunks.
- Generate vector embeddings for the processed knowledge.
- Store embeddings with metadata such as topic/source in a vector database.
- Accept employee questions through the assistant interface.
- Retrieve relevant knowledge using vector search.
- Generate an answer using the retrieved company knowledge.
- Return the relevant source/topic where possible.
- If information is unavailable, return a clear information-not-available fallback instead of inventing an answer.
- Escalate sensitive or appropriate queries to an authorized human when required.
- Handle workflow/API failures and protect credentials and sensitive information.

High-Level Architecture

Company Knowledge → Ingestion → Cleaning/Chunking → Embeddings → Vector DB → User Question
→ Retrieval → AI Generation → Source-Aware Response / Safe Fallback

4. Success Criteria

- At least 20 company knowledge entries are available in the knowledge base.
- Knowledge entries are successfully converted into embeddings and stored in the vector database.
- The assistant retrieves relevant knowledge for supported questions.
- Responses are grounded in retrieved company knowledge.
- Responses provide the relevant source/topic where possible.
- Questions outside the knowledge base receive a clear information-not-available fallback.
- The system does not fabricate company policies or internal information.
- Sensitive or appropriate queries can be escalated to a human.
- Errors are handled and logged without exposing credentials or unnecessary sensitive information.
- Credentials and secrets are not hard-coded or committed to the project.
- The solution can be demonstrated using normal, edge, and failure test cases.
- The project is documented sufficiently for another person to understand its setup and operation.

Project Status: Requirement analysis completed. The next stage is solution design and implementation.